Supplementary C — Dataset descriptors and tokenisation

Companion to: *A Band-Mask Robustness Diagnostic for LDA on HSI*

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A.1 Scope

This supplement expands Section III of the conference paper. We document per-scene the sensor history, band selection, ground-truth class catalogue, and the sampling decisions that produced the canonical 220-per-class corpora used in the band-mask diagnostic.

B.2 Per-scene sensor and corpus traceability

Table 1: Labelled HSI scenes used in the diagnostic. B is the band count after the canonical correction (uncorrected bands removed per the UPV/EHU collection's official policy). D is the document count that enters the canonical fit (= samples per class × number of admissible classes, with classes having fewer than 100 available pixels rejected). K is the canonical topic count.

Scene	Sensor / platform	B	D	K	Source URL
Indian Pines	AVIRIS / NASA	200	2812	12	ehu.eus/ccwintco/index.php?title=Hyperspectral
Salinas	AVIRIS / NASA	204	3520	12	ehu.eus / Salinas
Salinas-A	AVIRIS / NASA	204	1320	6	ehu.eus / Salinas-A (subset)
Pavia U	ROSIS-03 / DLR	103	1980	9	ehu.eus / Pavia University
KSC	AVIRIS / NASA	176	2686	12	ehu.eus / Kennedy Space Center
Botswana	EO-1 Hyperion / NASA	145	2775	12	ehu.eus / Botswana

B.2.1 Indian Pines

Sensor. Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), 224 raw bands at ≈ 10 nm resolution over 400–2500 nm. The canonical correction removes 24 noisy and atmospheric-water bands, leaving $B = 200$ usable bands. **Scene.** 145×145 pixels acquired over the agricultural test site at Purdue University, Indiana, June 1992 [1]. **Classes.** 16 ground-truth classes covering corn-notill, corn-mintill, corn, grass-pasture, grass-trees, hay-windrowed, oats, soybean-notill, soybean-mintill, soybean-clean, wheat, woods, buildings-grass-trees-drives, stone-steel-towers, alfalfa, grass-pasture-mowed. Some classes have fewer than 100 labelled pixels and are excluded from the canonical sample. **Sampling.** For each admissible class, the canonical fit draws 220 pixels without replacement using NumPy's `default_rng(42)`.

B.2.2 Salinas

Sensor. AVIRIS, same instrument family as Indian Pines. **Scene.** 512×217 pixels acquired over Salinas Valley, California [2], $B = 204$ usable bands after the canonical correction. **Classes.** 16 classes: brocoli-green-weeds-1/2, fallow, fallow-rough-plow, fallow-smooth, stubble, celery, grapes-untrained, soil-vinyard-develop, corn-senesced-green-weeds, lettuce-romaine-4,5,6,7wk, vinyard-untrained, vinyard-vertical-trellis.

B.2.3 Salinas-A

Sensor + classes. A subset of Salinas comprising 6 of the classes on a 83×86 window. Sensor history identical to Salinas. We treat it as an independent benchmark because its smaller spatial footprint and fewer classes make the canonical fit substantially easier than the parent scene.

B.2.4 Pavia University

Sensor. Reflective Optics System Imaging Spectrometer (ROSIS-03), DLR. 115 raw bands over 430–860 nm; the canonical correction removes 12 noisy bands leaving $B = 103$, all in the VNIR range. The scene is consequently *auto-skipped* for the SWIR mask in the diagnostic ($V^{(m)} = 0$ bands kept). **Scene.** 610×340 pixels over the University of Pavia, Italy [3]. **Classes.** 9 classes: asphalt, meadows, gravel, trees, painted-metal-sheets, bare-soil, bitumen, self-blocking-bricks, shadows.

B.2.5 Kennedy Space Center (KSC)

Sensor. AVIRIS-Old (AVIRIS recording before 1996). **Scene.** 512×614 pixels acquired over the Kennedy Space Center, Florida, March 1996. $B = 176$ usable bands after the canonical correction [4]. **Classes.** 13 classes covering wetland vegetation (salt-marsh, mud-flat, water, swamp, etc.). **Note.** KSC is the labelled scene most frequently observed to yield a band-fragile canonical basis, an observation surfaced by the diagnostic; the supplementary E document discusses possible mechanisms in more depth.

B.2.6 Botswana

Sensor. Earth Observing 1 (EO-1) Hyperion. **Scene.** 1476×256 pixels acquired over the Okavango Delta, Botswana, May 2001 [5]. $B = 145$ usable bands after the canonical correction. **Classes.** 14 classes including water, hippo-grass, floodplain-grasses-1/2, reeds, riparian, firescar-2, island-interior, acacia-woodlands, acacia-shrublands, acacia-grasslands, short-mopane, mixed-mopane, exposed-soils.

C.3 Tokenisation reproducibility note

The min-max normalisation of §2.1 of supplementary B is computed *within each pixel* and not across the corpus, so the tokenisation of pixel d depends only on the spectrum of pixel d. This is by design: across-corpus normalisation would couple the tokenisation to the sample selection, and the band-mask diagnostic would then conflate spectral-window restriction with corpus-extent sensitivity. The trade-off is that pixels with very flat spectra (e.g. heavy shadow) tokenise to a uniform count vector, which the LDA fit downweights via the small α prior.

D.4 HIDSAG mineral subsets

The five HIDSAG subsets evaluated alongside the labelled scenes are documented in detail in the journal supplement E. Briefly: HIDSAG is a curated collection of point-spectrum measurements over mineralogical samples acquired with a SWIR-only sensor; the five subsets (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY) collectively cover 19 target labels distributed across multiple samples. The band-mask diagnostic on HIDSAG uses *between-covariate-variance* band ranking in place of the labelled-scene Fisher discriminant, as HIDSAG has no per-pixel ground truth.

References

[1] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe, "220 band AVIRIS hyperspectral image data set: June 12, 1992 Indian Pine test site 3," Purdue University Research Repository, 2015, indian Pines AVIRIS hyperspectral dataset, calibrated 220-band reflectance, ground-truth labels for 16 land-cover classes.

[2] NASA/JPL Airborne Visible/Infrared Imaging Spectrometer Project, "AVIRIS hyperspectral imagery — Salinas valley, california," https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Salinas_scene, 1998, 224-band hyperspectral cube, 3.7 m spatial resolution. Calibrated and corrected version maintained by University of Pavia GIC. Salinas-A is a 86×83 sub-scene with 6 classes.

[3] German Aerospace Center (DLR) ROSIS sensor, "ROSIS Pavia University hyperspectral dataset," https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University, 2003, reflective Optics System Imaging Spectrometer airborne acquisition over Pavia, Italy. 103 bands after bad-band removal, 9 classes, 610×340 pixels at 1.3 m resolution.

[4] NASA/JPL Airborne Visible/Infrared Imaging Spectrometer Project, "AVIRIS Kennedy Space Center hyperspectral imagery," https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Kennedy_Space_Center_.28KSC.29, 1996, kennedy Space Center, Florida, 18 m resolution, 224 bands, 13 land-cover classes after labelling.

[5] NASA EO-1 Hyperion Mission, "EO-1 Hyperion Okavango Botswana hyperspectral imagery," https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Botswana, 2001, hyperion satellite-borne hyperspectral sensor, Okavango Delta, Botswana. 145 bands after preprocessing, 14 land-cover classes.